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Jeanneret

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(54) **LADDER SAFETY APPARATUS**

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CPC ***E06C 7/482*** (2013.01); ***E06C 1/34*** (2013.01); *E06C 1/12* (2013.01)

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CPC ... E06C 1/34; E06C 7/081; E06C 7/48; E06C 7/482; E06C 7/16; E06C 7/165; A01M 31/02
See application file for complete search history.

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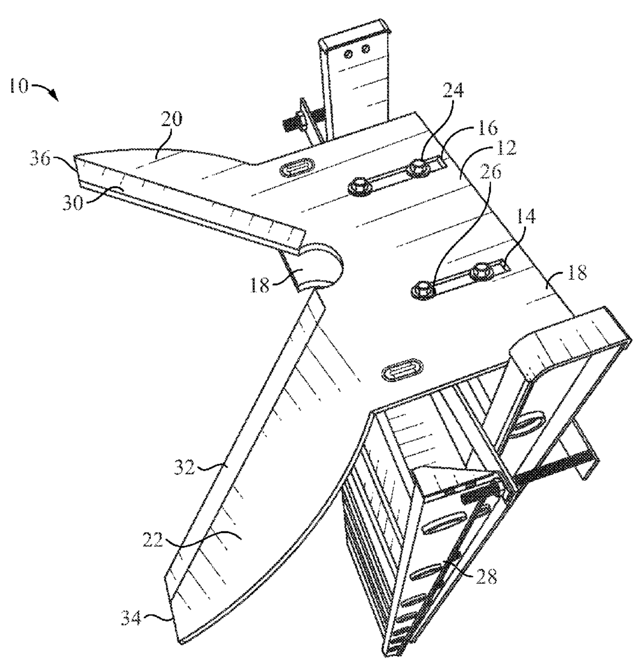
(57) **ABSTRACT**

A ladder attachment may allow a ladder to securely rest on a corner of a structure, structure corners, tree, or the like. The attachment may be attached to the rung of a ladder via several mounting brackets and mounting flanges. The attachment includes first and second arms which may be arranged at about a 90° angle to allow the positioning a ladder on the corner of the structure or oddly shaped structures during use.

20 Claims, 4 Drawing Sheets



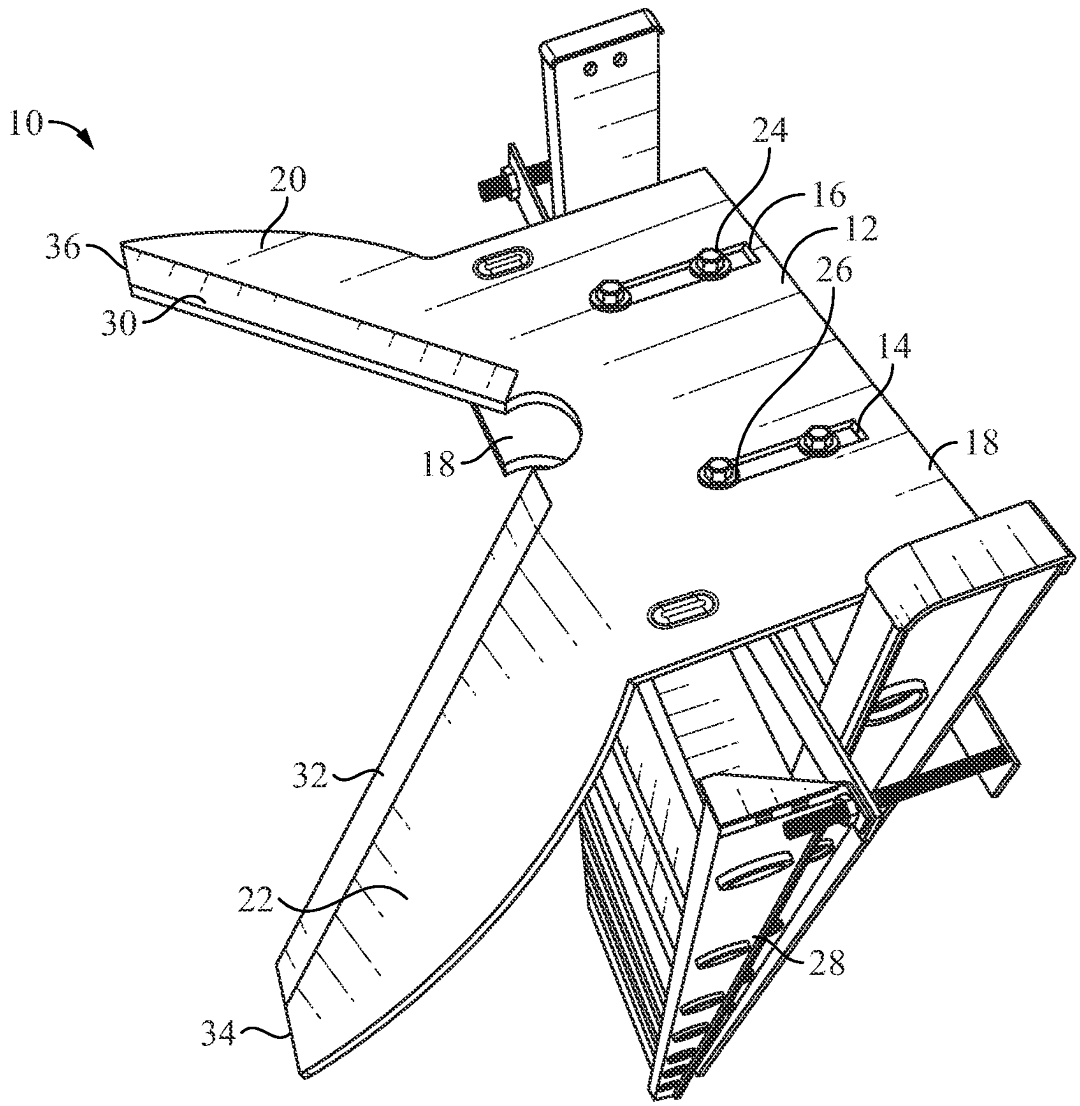


FIG. 1

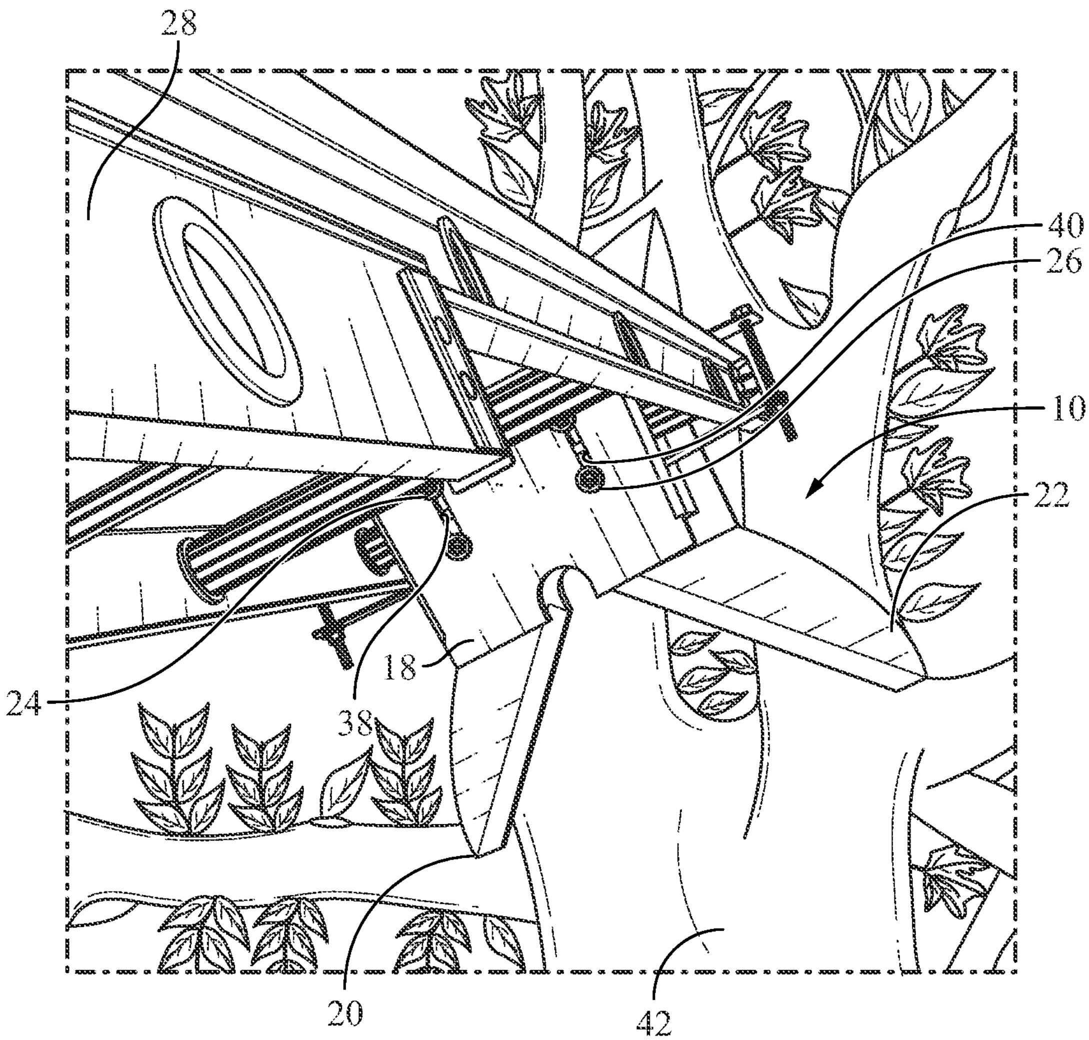


FIG. 2

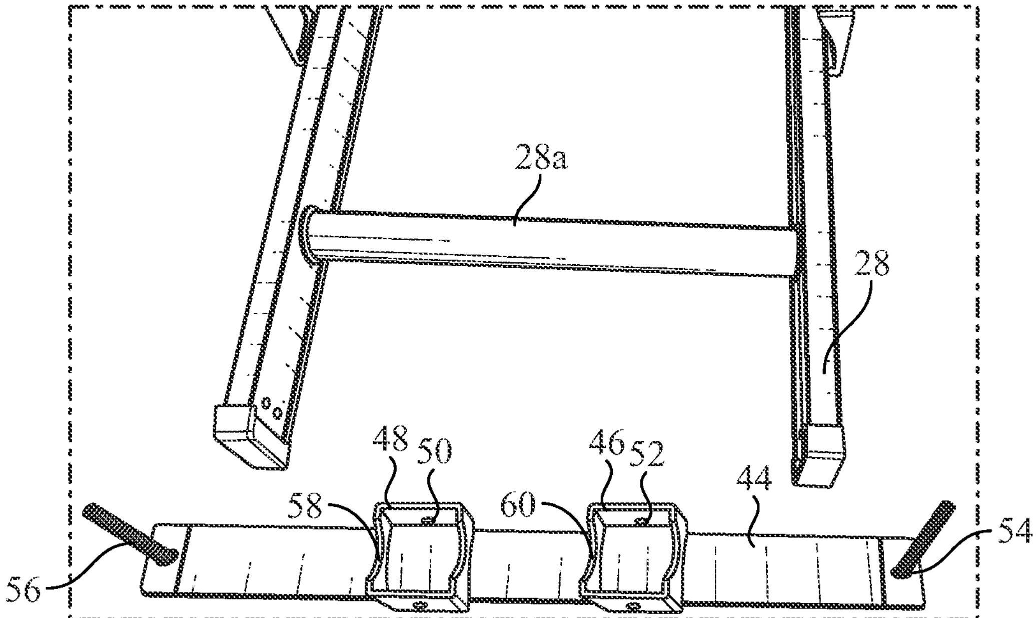


FIG. 3A

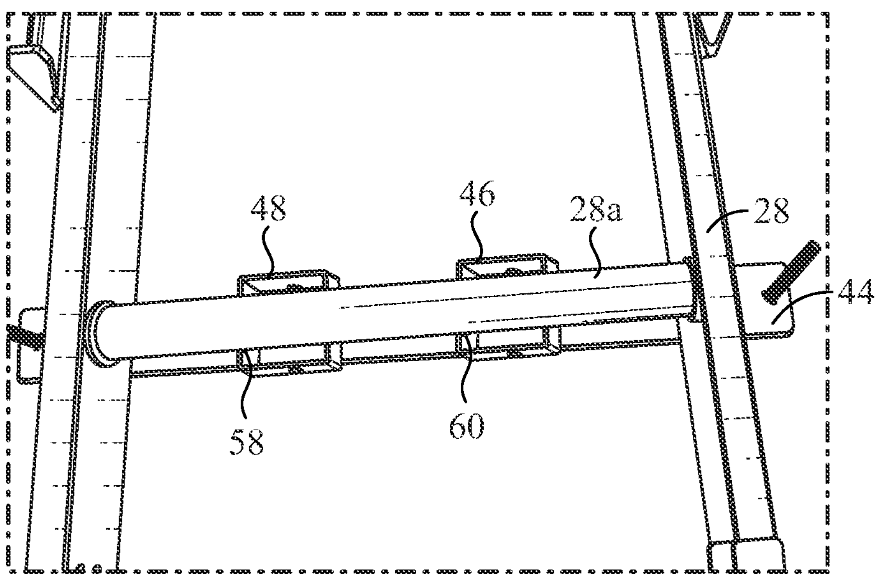


FIG. 3B

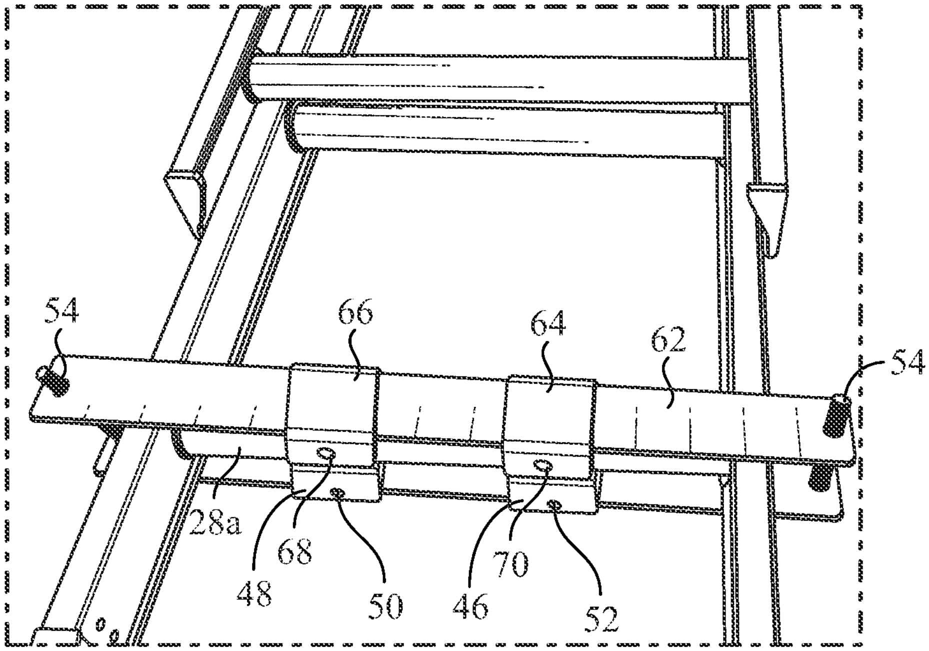

54
66
64
62
54
28a
68
48
50
46
70
52

FIG. 3C

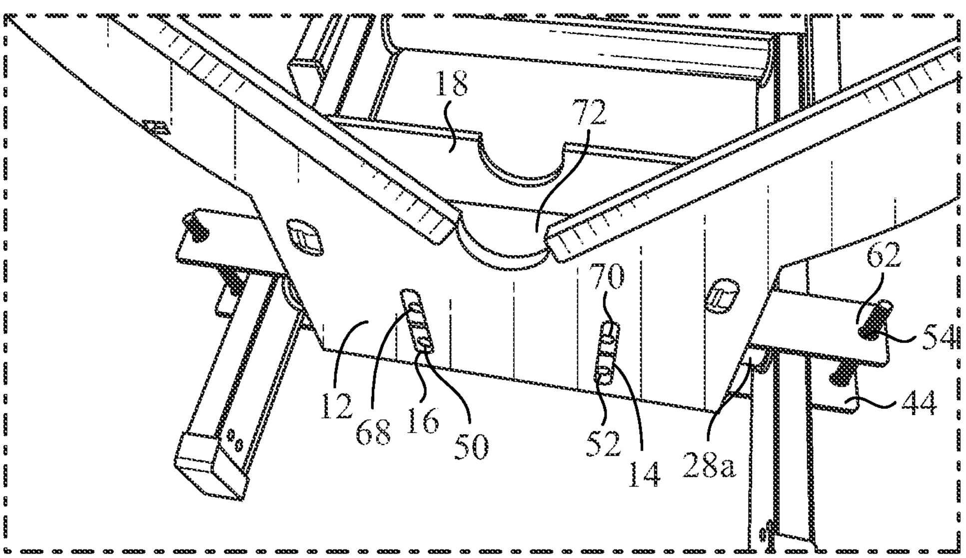

18
72
70
62
54
44
12
68
16
50
52
14
28a

FIG. 3D

LADDER SAFETY APPARATUS

TECHNICAL FIELD

The embodiments generally relate to the field of ladder safety.

BACKGROUND

Ladders may be difficult to safely lean or prop against corners or round features on structures or trees. Reducing ladder accidents, and in particular those involving dangerous lateral movement of the ladder, by providing a means of embracing existing corners and round features on structures or trees can improve overall safety of ladder use.

SUMMARY

This summary is provided to introduce a variety of concepts in a simplified form that is further disclosed in the detailed description of the embodiments. This summary is not intended to identify key or essential inventive concepts of the claimed subject matter, nor is it intended for determining the scope of the claimed subject matter.

A ladder attachment may allow a ladder to securely rest on a corner of a structure, tree, or the like. The attachment releasably engages via a plurality of bolts to the top end of a ladder and extends laterally to contact the corner of the structure. The attachment includes first and second arms which may be arranged at about a 90° angle to allow the ladder to securely rest on the corner of a structure during use. A central portion allows for the ladder to securely rest on a structure corner, tree, or the like.

The ladder attachment may include a top panel affixed to a bottom panel via a web similar to that of a traditional I-beam. The top panel may be constructed and arranged to engage and securely rest against a corner of a structure, tree, or various other structures.

Other illustrative variations within the scope of the invention will become apparent from the detailed description provided hereinafter. The detailed description and enumerated variations, while disclosing optional variations, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

A complete understanding of the present embodiments and the advantages and features thereof will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:

FIG. **1** illustrates a view of an assembled ladder attachment according to some variations described herein;

FIG. **2** illustrates a view of an assembled ladder attachment according to some variations described herein;

FIG. **3***a* illustrates a view of a portion of a ladder attachment according to some variations described herein;

FIG. **3***b* illustrates a view of a portion of a ladder attachment according to some variations described herein;

FIG. **3***c* illustrates a view of a portion of a ladder attachment according to some variations described herein; and

FIG. **3***d* illustrates a view of a portion of a ladder attachment according to some variations described herein.

DETAILED DESCRIPTION

The specific details of the single embodiment or variety of embodiments described herein are to the described system and methods of use. Any specific details of the embodiments are used for demonstration purposes only and no unnecessary limitations or inferences are to be understood from there.

It is noted that the embodiments reside primarily in combinations of components and procedures related to the system. Accordingly, the system components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

The ladder attachment may include a top panel affixed to a bottom panel via a web. The top panel may be constructed and arranged to engage and securely rest against a corner of a structure, tree, or various other structures via first and second arms extending away from the ladder attachment. The ladder attachment may include a first and second mounting brackets constructed and arranged to mechanically connect to the top and bottom panels to secure the ladder attachment to a rung of a ladder.

The scope of the invention will be best understood by the following description of figures. FIG. **1** shows a top-down perspective view of a variation of a ladder attachment **10** that may include a top plate **12** defining a first slot **14** and second slot **16**. The top plate **12** may include a first arm **20** and a second arm **22** extending therefrom wherein the first arm **20** and the second arm **22** extend from the top plate **12** at about a 90-degree angle from one another. The arrangement of the first arm **20** and second arm **22** may include narrower or wider angles with respect to position between the two. The first arm **20** may include a generally blade shaped profile **36** constructed and arranged to engage a corner of a structure, tree, or the like. The first arm **20** may include a liner **30** constructed and arranged to pad the ladder attachment **10** when engaged with a corner of a structure, tree, or the like to reduce damage and wear between the first arm **20** and the corner or tree. The liner **30** may include adhesive tape, foam, padding, or the like. The second arm **22** may include a generally blade shaped profile **34** constructed and arranged to engage a corner of a structure, tree, or the like. The second blade **22** may include a liner **32** constructed and arranged to pad the ladder attachment **10** when engaged with a corner, tree, or the like to reduce damage and wear between the second arm **22** and the corner or tree. The liner **32** may include adhesive tape, foam, padding come over the like. The ladder attachment **10** may be connected to a rung of a ladder **28** by means of mounting brackets and a plurality of nut and bolt combination **24**, **26** passed through the first slot **14** and the second slot **16** and secured on the second plate **18**.

FIG. **2** shows a bottom-up perspective view of a variation of a ladder attachment **10** that may include a bottom plate **18** securely attaching the ladder attachment **10** to a rung of a ladder **28** via mounting brackets and a plurality of nut and bolt combinations **24**, **26**, passing through a third slot **38** and a fourth slot **40** defined by the bottom plate **18**. The first arm **20** and second arm **22** may be constructed and arranged to securely engage with a corner, tree, or the like **42**.

FIGS. **3***a*-**3***d* show multiple views of assembling a ladder attachment **10** to a ladder **28**. As seen in FIG. **3***a*, the ladder attachment may include a first mounting bracket **44** that may include a first mounting flange **46** and a second mounting flange **48** constructed and arranged to engage with a rung **28***a* of a ladder **28**. The flange may generally be a rectangular tube in shape or may be any other suitable cross section such that the flange **46** may engage a rung **28***a*. The first mounting flange **46** may define recess **60** constructed and arranged to engage with a rung **28***a*. The recess **60** may be generally semi-circular in shape or may be any other suitable shape such that the recess **60** may receive a portion of a rung **28***a*. The first mounting flange **46** may additionally define through holes **52** constructed and arranged to receive a plurality of nut and bolt combinations **24**, **26** when the ladder attachment is fully assembled. Similarly, the second mounting flange **48** may define recess **58** constructed and arranged to engage with a rung **28***a*. the second mounting flange **48** may additionally defined through holes **50** constructed and arranged to receive the plurality of nut and bolt combinations **24**, **26** when the ladder attachment is fully assembled. The mounting bracket **44** may further include a second plurality of nut and bolt combinations **54**, **56** to facilitate mounting the first mounting bracket **44** to a second mounting bracket as best seen in FIG. **3***c*.

Referring to FIG. **3***b*, the first mounting bracket **44** may be placed under the rung **28** of a ladder **28** such that the rung **28***a* is seated within the recess **60** of the first mounting flange **46** and the recess **58** of the second mounting flange **48**.

Referring to FIG. **3***c*, a second mounting bracket **62** may be placed on the rung **28***a* such that the third mounting flange **64** and the fourth mounting flange **66** may be seated on the rung **28***a* similar to that of the first mounting flange **46** and the second mounting flange **48** such that the rung **28***a* is mechanically clamped between the first mounting bracket **44** and the second mounting bracket **62**. The second plurality of nut and bolt combinations **54**, **56** may passed through a portion of the second mounting bracket **62** and may be constructed and arranged to secure the first mounting bracket **44** and the second mounting bracket **62** on the rung **28***a*. The third mounting flange **64** may additionally defined through holes **70** constructed and arranged to receive the plurality of nut and bolt combinations **24**, **26** when the ladder attachment is fully assembled. The fourth mounting flange **66** may additionally define through holes **68** constructed and arranged to receive the plurality of nut and bolt combinations **24**, **26** when the ladder attachment is fully assembled.

Referring to FIG. **3***d*, the top plate **12**, web **72**, and bottom plate **18** may be positioned over the secured first mounting bracket **44** and second mounting bracket **62** and the second plurality of nut and bolt combinations **54** and **56** may be passed through the first slot **14**, the second slot **16**, third slot **38**, fourth slot **40**, and through holes **50**, **52**, **68**, and **70** such that the ladder attachment is secured on the rung **28***a*.

The following description of variants is only illustrative of components, elements, acts, products, and methods considered to be within the scope of the invention and are not in any way intended to limit such scope by what is specifically disclosed or not expressly set forth. The components, elements, acts, products, and methods as described herein may be combined and rearranged other than as expressly described herein and are still considered to be within the scope of the invention.

According to variation 1, a ladder safety apparatus may include a top plate including a body, a first arm, and a second arm, the first arm and second arm extending away from the body of the top plate approximately perpendicular to one another, the top plate defining a first through hole and a second through hole therein. The ladder safety apparatus may include a bottom plate defining a third through hole and a fourth through hole therein; and a web connecting the body of the top plate to the bottom plate. The ladder safety apparatus may include a first mounting bracket including a first mounting flange and a second mounting flange each constructed and arranged to engage with a rung of a ladder, the first mounting bracket defining a fifth through hole and a sixth through hole. The ladder safety apparatus may include a second mounting bracket including a third mounting flange and a fourth mounting flange each constructed and arranged to engage with a rung of a ladder, the second mounting bracket defining a seventh through hole and an eighth through hole. The ladder safety apparatus may include a first plurality of nut and bolt combinations constructed and arranged to thread through the fifth, sixth, seventh, and eighth through holes to mechanically connect the first mounting bracket to the second mounting bracket such that a rung of a ladder is engaged by the first, second, third, and fourth mounting flanges. The top plate, web, and bottom plate are constructed and arranged to mechanically connect to the mechanically connected first mounting bracket and second mounting bracket such that the top plate, web, and bottom plate are secured to a rung of a ladder.

Variation 2 may include a ladder safety apparatus as in variation 1, wherein the top plate and bottom plate are approximately parallel to each other and the web is connected approximately perpendicular to both the top plate and bottom plate;

Variation 3 may include a ladder safety apparatus as in any of variations 1 through 2, wherein the top plate, bottom plate, and web include aluminum.

Variation 4 may include a ladder safety apparatus as in any of variations 1 through 3, wherein the first, second, third, and fourth mounting flanges are rectangular tube in shape.

Variation 5 may include a ladder safety apparatus as in any of variations 1 through 4, wherein the first, second, third, and fourth mounting flanges define respective first, second, third, and fourth recesses therein constructed and arranged to engage with a rung of a ladder.

Variation 6 may include a ladder safety apparatus as in any of variations 1 through 5, wherein the first, second, third, and fourth recesses are approximately semicircular in shape.

Variation 7 may include a ladder safety apparatus as in any of variations 1 through 6, further including a liner disposed on the first arm and second arm.

Variation 8 may include a ladder safety apparatus as in any of variations 1 through 7, wherein the first and the second arm are generally blade shaped.

Variation 9 may include a ladder safety apparatus as in any of variations 1 through 8, wherein the top plate defines an arc-shaped gap between the first arm and the second arm.

According to variation 10, a ladder safety apparatus may include a top plate including a body, a first arm, and a second arm, the first arm and second arm extending away from the body of the top plate approximately perpendicular to one another, the top plate defining a first through hole and a second through hole therein. The ladder safety apparatus may include a bottom plate defining a third through hole and a fourth through hole therein and a web connecting the body of the top plate to the bottom plate. The ladder safety apparatus may include a first mounting bracket defining a fifth through hole and a sixth through hole and a first mounting flange connected to the first mounting bracket, the first mounting flange defining a first set of through holes and a first recess constructed and arranged to engage a rung of a ladder. The ladder safety apparatus may include a second mounting flange connected to the first mounting bracket, the second mounting flange defining a second set of through holes and a second recess constructed and arranged to engage a rung of a ladder. The ladder safety apparatus may include a second mounting bracket defining a seventh through hole and an eighth through hole and a third mounting flange connected to the second mounting bracket, the third mounting flange defining a third set of through holes and a third recess constructed and arranged to engage a rung of a ladder. The ladder safety apparatus may include a fourth mounting flange connected to the second mounting bracket, the fourth mounting flange defining a fourth set of through holes and a fourth recess constructed and arranged to engage a rung of a ladder. The ladder safety apparatus may include a first plurality of nut and bolt combinations constructed and arranged to thread through the fifth, sixth, seventh, and eighth through holes to mechanically connect the first mounting bracket to the second mounting bracket such that a rung of a ladder is engaged by the first, second, third, and fourth mounting flanges. The ladder safety apparatus may include a second plurality of nut and bolt combinations constructed and arranged to thread through the first, second, third, and fourth through holes, and through the first, second, third, and fourth sets of through holes such that the top plate, web, and bottom plate may be connected to the first and second mounting brackets.

Variation 11 may include a ladder safety apparatus as in variation 10, wherein the top plate and bottom plate are approximately parallel to each other and the web is connected approximately perpendicular to both the top plate and bottom plate;

Variation 12 may include a ladder safety apparatus as in any of variations 10 through 11, wherein the top plate, bottom plate, and web include aluminum.

Variation 13 may include a ladder safety apparatus as in any of variations 10 through 12, wherein the first, second, third, and fourth mounting flanges are rectangular tube in shape.

Variation 14 may include a ladder safety apparatus as in any of variations 10 through 13, wherein the first, second, third, and fourth recesses are approximately semicircular in shape.

Variation 15 may include a ladder safety apparatus as in any of variations 10 through 14, further including a liner disposed on the first arm and second arm.

Variation 16 may include a ladder safety apparatus as in any of variations 10 through 15, wherein the first and the second arm are generally blade shaped.

Variation 17 may include a ladder safety apparatus as in any of variations 10 through 16, wherein the top plate defines an arc-shaped gap between the first arm and the second arm.

According to variation 18, a ladder safety apparatus may include a ladder including at least one rung and a top plate including a body, a first arm, and a second arm, the first arm and second arm extending away from the body of the top plate approximately perpendicular to one another, the top plate defining a first through hole, a second through hole, and an arc-shaped gap between the first arm and the second arm therein. The ladder safety apparatus may include a liner disposed on the first arm and the second arm; a bottom plate defining a third through hole and a fourth through hole therein; and a web connecting the body of the top plate to the bottom plate. The ladder safety apparatus may include a first mounting bracket defining a fifth through hole and a sixth through hole and a first mounting flange connected to the first mounting bracket, the first mounting flange defining a first set of through holes and a first recess constructed and arranged to engage the at least one rung of the ladder. The ladder safety apparatus may include a second mounting flange connected to the first mounting bracket, the second mounting flange defining a second set of through holes and a second recess constructed and arranged to engage the at least one rung of the ladder. The ladder safety apparatus may include a second mounting bracket defining a seventh through hole and an eighth through hole and a third mounting flange connected to the second mounting bracket, the third mounting flange defining a third set of through holes and a third recess constructed and arranged to engage the at least one rung of the ladder. The ladder safety apparatus may include a fourth mounting flange connected to the second mounting bracket, the fourth mounting flange defining a fourth set of through holes and a fourth recess constructed and arranged to engage the at least one rung of the ladder. The ladder safety apparatus may include a first plurality of nut and bolt combinations constructed and arranged to thread through the fifth, sixth, seventh, and eighth through holes to mechanically connect the first mounting bracket to the second mounting bracket such that the at least one rung of the ladder is engaged by the first, second, third, and fourth mounting flanges. The ladder safety apparatus may include a second plurality of nut and bolt combinations constructed and arranged to thread through the first, second, third, and fourth through holes, and through the first, second, third, and fourth sets of through holes such that the top plate, web, and bottom plate may be connected to the first and second mounting brackets.

Variation 19 may include a ladder safety apparatus as in variation 18, wherein the first, second, third, and fourth recesses are approximately semicircular in shape.

Variation 20 may include a ladder safety apparatus as in any of variations 18 through 19, wherein the top plate and bottom plate are approximately parallel to each other and the web is connected approximately perpendicular to both the top plate and bottom plate.

Many different embodiments have been disclosed herein, in connection with the above description and the drawing. It will be understood that it would be unduly repetitious and obfuscating to describe and illustrate every combination and subcombination of these embodiments. Accordingly, all embodiments can be combined in any way and/or combination, and the present specification, including the drawings, shall be construed to constitute a complete written description of all combinations and subcombinations of the embodiments described herein, and of the manner and process of making and using them, and shall support claims to any such combination or subcombination.

An equivalent substitution of two or more elements can be made for anyone of the elements in the claims below or that a single element can be substituted for two or more elements in a claim. Although elements can be described above as acting in certain combinations, and even initially claimed as such, it is to be expressly understood that one or more elements from a claimed combination can, in some cases, be excised from the combination and that the claimed combination can be directed to a subcombination or variation of a subcombination.

It will be appreciated by persons skilled in the art that the present embodiment is not limited to what has been particularly shown and described hereinabove. A variety of modifications and variations are possible considering the above teachings without departing from the following claims.

What is claimed is:

1. A ladder safety apparatus, comprising:

a top plate comprising a body, a first arm, and a second arm, the first arm and second arm extending away from the body of the top plate approximately perpendicular to one another, the top plate defining a first through hole and a second through hole therein;

a bottom plate defining a third through hole and a fourth through hole therein; a web connecting the body of the top plate to the bottom plate;

a first mounting bracket comprising a first mounting flange and a second mounting flange each constructed and arranged to engage with a rung of a ladder, the first mounting bracket defining a fifth through hole and a sixth through hole;

a second mounting bracket comprising a third mounting flange and a fourth mounting flange each constructed and arranged to engage with a rung of a ladder, the second mounting bracket defining a seventh through hole and an eighth through hole;

a first plurality of nut and bolt combinations constructed and arranged to thread through the fifth, sixth, seventh, and eighth through holes to mechanically connect the first mounting bracket to the second mounting bracket such that a rung of a ladder is engaged by the first, second, third, and fourth mounting flanges; and

wherein the top plate, web, and bottom plate are constructed and arranged to mechanically connect to the mechanically connected first mounting bracket and second mounting bracket such that the top plate, web, and bottom plate are secured to a rung of a ladder.

2. A ladder safety apparatus as in claim **1**, wherein the top plate and bottom plate are approximately parallel to each other and the web is connected approximately perpendicular to both the top plate and bottom plate.

3. A ladder safety apparatus as in claim **1**, wherein the top plate, bottom plate, and web comprise aluminum.

4. A ladder safety apparatus as in claim **1**, wherein the first, second, third, and fourth mounting flanges are rectangular tube in shape.

5. A ladder safety apparatus as in claim **1**, wherein the first, second, third, and fourth mounting flanges define respective first, second, third, and fourth recesses therein constructed and arranged to engage with a rung of a ladder.

6. A ladder safety apparatus as in claim **5**, wherein the first, second, third, and fourth recesses are approximately semicircular in shape.

7. A ladder safety apparatus as in claim **1**, further comprising a liner disposed on the first arm and second arm.

8. A ladder safety apparatus as in claim **1**, wherein the first and the second arm are generally blade shaped.

9. A ladder safety apparatus as in claim **1**, wherein the top plate defines an arc-shaped gap between the first arm and the second arm.

10. A ladder safety apparatus, comprising:

a top plate comprising a body, a first arm, and a second arm, the first arm and second arm extending away from the body of the top plate approximately perpendicular to one another, the top plate defining a first through hole and a second through hole therein;

a bottom plate defining a third through hole and a fourth through hole therein;

a web connecting the body of the top plate to the bottom plate;

a first mounting bracket defining a fifth through hole and a sixth through hole;

a first mounting flange connected to the first mounting bracket, the first mounting flange defining a first set of through holes and a first recess constructed and arranged to engage a rung of a ladder;

a second mounting flange connected to the first mounting bracket, the second mounting flange defining a second set of through holes and a second recess constructed and arranged to engage a rung of a ladder;

a second mounting bracket defining a seventh through hole and an eighth through hole;

a third mounting flange connected to the second mounting bracket, the third mounting flange defining a third set of through holes and a third recess constructed and arranged to engage a rung of a ladder;

a fourth mounting flange connected to the second mounting bracket, the fourth mounting flange defining a fourth set of through holes and a fourth recess constructed and arranged to engage a rung of a ladder;

a first plurality of nut and bolt combinations constructed and arranged to thread through the fifth, sixth, seventh, and eighth through holes to mechanically connect the first mounting bracket to the second mounting bracket such that a rung of a ladder is engaged by the first, second, third, and fourth mounting flanges; and

a second plurality of nut and bolt combinations constructed and arranged to thread through the first, second, third, and fourth through holes, and through the first, second, third, and fourth sets of through holes such that the top plate, web, and bottom plate may be connected to the first and second mounting brackets.

11. A ladder safety apparatus as in claim **10**, wherein the top plate and bottom plate are approximately parallel to each other and the web is connected approximately perpendicular to both the top plate and bottom plate.

12. A ladder safety apparatus as in claim **10**, wherein the top plate, bottom plate, and web comprise aluminum.

13. A ladder safety apparatus as in claim **10**, wherein the first, second, third, and fourth mounting flanges are rectangular tube in shape.

14. A ladder safety apparatus as in claim **10**, wherein the first, second, third, and fourth recesses are approximately semicircular in shape.

15. A ladder safety apparatus as in claim **10**, further comprising a liner disposed on the first arm and second arm.

16. A ladder safety apparatus as in claim **10**, wherein the first and the second arm are generally blade shaped.

17. A ladder safety apparatus as in claim **10**, wherein the top plate defines an arc-shaped gap between the first arm and the second arm.

18. A ladder safety apparatus, comprising:

a ladder comprising at least one rung;

a top plate comprising a body, a first arm, and a second arm, the first arm and second arm extending away from the body of the top plate approximately perpendicular to one another, the top plate defining a first through hole, a second through hole, and an arc-shaped gap between the first arm and the second arm therein;

a liner disposed on the first arm and the second arm;

a bottom plate defining a third through hole and a fourth through hole therein;

a web connecting the body of the top plate to the bottom plate;

a first mounting bracket defining a fifth through hole and a sixth through hole;

a first mounting flange connected to the first mounting bracket, the first mounting flange defining a first set of through holes and a first recess constructed and arranged to engage the at least one rung of the ladder;

a second mounting flange connected to the first mounting bracket, the second mounting flange defining a second set of through holes and a second recess constructed and arranged to engage the at least one rung of the ladder;

a second mounting bracket defining a seventh through hole and an eighth through hole;

a third mounting flange connected to the second mounting bracket, the third mounting flange defining a third set of through holes and a third recess constructed and arranged to engage the at least one rung of the ladder;

a fourth mounting flange connected to the second mounting bracket, the fourth mounting flange defining a fourth set of through holes and a fourth recess constructed and arranged to engage the at least one rung of the ladder;

a first plurality of nut and bolt combinations constructed and arranged to thread through the fifth, sixth, seventh, and eighth through holes to mechanically connect the first mounting bracket to the second mounting bracket such that the at least one rung of the ladder is engaged by the first, second, third, and fourth mounting flanges; and

a second plurality of nut and bolt combinations constructed and arranged to thread through the first, second, third, and fourth through holes, and through the first, second, third, and fourth sets of through holes such that the top plate, web, and bottom plate may be connected to the first and second mounting brackets.

19. A ladder safety apparatus as in claim **18**, wherein the first, second, third, and fourth recesses are approximately semicircular in shape.

20. A ladder safety apparatus as in claim **18**, wherein the top plate and bottom plate are approximately parallel to each other and the web is connected approximately perpendicular to both the top plate and bottom plate.

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